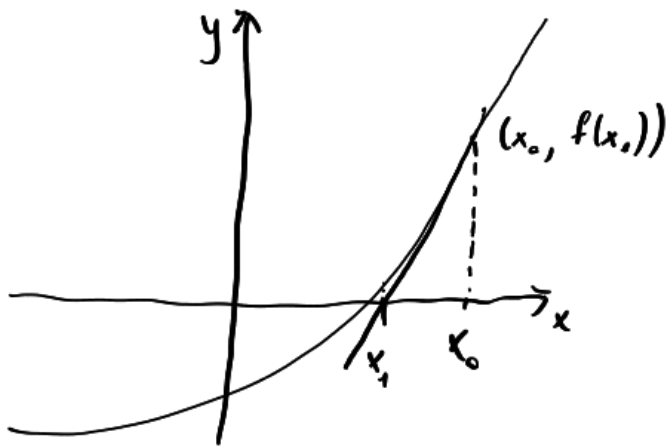


1a. Z uporabo Newt. metode poišči rešitev

$$x-1 = \frac{1}{2} \sin x \text{ na 3 decimalke natančno}$$

Newtonova metoda:



$$f(x) = 0:$$

① tangenta v točki  $x_0 = kx + n$

②  $kx + n = 0 \rightarrow x_1$

③ ponovljamo

$$y - f(x_0) = f'(x_0)(x - x_0)$$

$$\downarrow y=0$$

$$0 = f'(x_0)x + f(x_0) - x_0 f'(x_0)$$

$$\downarrow$$

$$x_1: x = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\downarrow$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x) = x - 1 - \frac{1}{2} \sin x$$

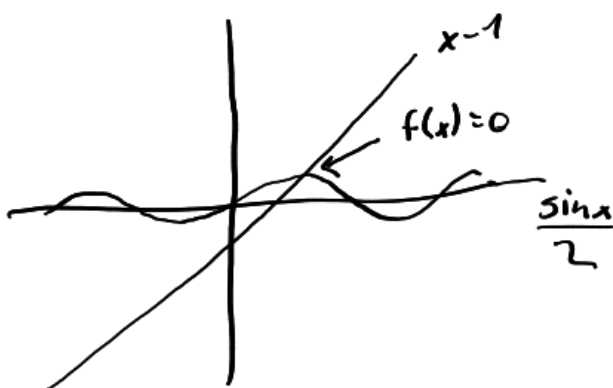
$$f'(x) = 1 - \frac{1}{2} \cos x$$

$$x_0 = 2$$

$$x_1 = 2 - \frac{1 - \frac{1}{2} \sin 2}{1 - \frac{1}{2} \cos 2} = 1,54858$$

$$x_2 = 1,49933$$

$$(x_3 = 1,4987)$$



1.) Kesi 2. uveliko hole spreminjamo

$$a) \int \frac{1}{x^2+9} dx = \frac{1}{9} \int \frac{1}{1+(\frac{x}{3})^2} dx = \frac{1}{9} \int \frac{3}{1+t^2} dt = \boxed{\frac{1}{3} \arctg \frac{x}{3} + C}$$

$$\frac{x}{3} = t$$

$$\frac{1}{3} dx = dt$$

$$b) \int \frac{1}{\sqrt{a^2-x^2}} dx = \frac{1}{a} \int \frac{dx}{\sqrt{1-(\frac{x}{a})^2}} = \boxed{\arcsin \frac{x}{a} + C}$$

$a > 0$

$$c) \int \sin^3 x \cos x dx = \int t^3 dt = \boxed{\frac{\sin^4 x}{4} + C}$$

$$\sin x = t$$

$$\cos x dx = dt$$

$$d) \int \frac{2x}{x^2+25} dx = \int \frac{dt}{t} = \boxed{\ln(x^2+25) + C}$$

$$x^2+25 = t$$

$$2x dx = dt$$

ne rabi bit abs., ker je vedno  $> 0$

$$e) \int \frac{1}{\sqrt{1+e^{2x}}} dx = \int \frac{1}{t^2-1} dt = \frac{1}{2} \int \frac{1}{t-1} dt - \frac{1}{2} \int \frac{1}{t+1} dt =$$

$$= \frac{1}{2} \int \frac{1}{u} du - \frac{1}{2} \int \frac{1}{v} dv = \frac{1}{2} \ln|t-1| - \frac{1}{2} \ln|t+1| + C = \boxed{\frac{1}{2} \ln \frac{\sqrt{1+e^{2x}}-1}{\sqrt{1+e^{2x}}+1} + C}$$

$$t = \sqrt{1+e^{2x}} \rightarrow e^{2x} = t^2 - 1$$

$$\frac{1}{(t+1)(t-1)} = \frac{A}{t+1} + \frac{B}{t-1}$$

$$dt = \frac{1}{2\sqrt{1+e^{2x}}} \cdot 2e^{2x} dx$$

$$At - A + Bt + B = 1$$

$$B - A = 1$$

$$A = -B$$

$$t-1 = u$$

$$t+1 = v$$

$$dt = du$$

$$dt = du$$

$$2B = 1$$

$$B = \frac{1}{2}$$

$$A = -\frac{1}{2}$$

2. Laško partes:

$$\int u dv = u \cdot v - \int v du$$

a)  $\int \arctg x dx = x \arctg x - \int \frac{x}{x^2+1} dx = x \arctg x - \frac{1}{2} \int \frac{1}{t} dt =$

$$\begin{array}{lll} u = \arctg x & dv = dx & x^2+1 = t \\ du = \frac{1}{x^2+1} dx & v = x & 2x dx = dt \end{array}$$

$$= x \arctg x - \frac{1}{2} \ln|x^2+1| + C$$

b)  $\int \arcsin x dx = x \arcsin x - \int \frac{x}{\sqrt{1-x^2}} dx = x \arcsin x + \frac{1}{2} \int \frac{dt}{\sqrt{t}} =$

$$\begin{array}{lll} u = \arcsin x & dv = dx & 1-x^2 = t \\ du = \frac{1}{\sqrt{1-x^2}} dx & v = x & dt = -2x dx \end{array}$$

$$= x \arcsin x + \sqrt{1-x^2} + C$$

c)  $\int e^{ax} \sin bx dx = \frac{1}{a} e^{ax} \sin bx - \frac{b}{a} \int e^{ax} \cos bx dx =$   
 $a, b \in \mathbb{R}$

$$\begin{array}{ll} u = \sin bx & e^{ax} dx = dv \\ du = b \cos bx dx & \frac{1}{a} e^{ax} = v \end{array}$$

$$\int e^{ax} dx = \frac{1}{a} \int e^t dt = \frac{1}{a} e^{ax} + C$$

$$\begin{array}{l} ax = t \\ a dx = dt \end{array}$$

$$\begin{array}{ll} u' = \cos bx & dv' = e^{ax} dx \\ du' = -b \sin bx & v' = \frac{1}{a} e^{ax} \end{array}$$

$$= \frac{1}{a} e^{ax} \sin bx - \frac{1}{a} \left( \frac{1}{a} e^{ax} \cos bx + \frac{b}{a} e^{ax} \sin bx \right) = \frac{1}{a} e^{ax} \left( \sin bx - \frac{b}{a} \cos bx \right) - \frac{b}{a^2} e^{ax} \sin bx \dots$$

$$\left(1 + \frac{b^2}{a^2}\right) \int e^{ax} \sin bx dx = \frac{1}{a} e^{ax} \left( \sin bx - \frac{b}{a} \cos bx \right)$$

$$\boxed{\int e^{ax} \sin bx dx = \frac{ae^{ax} \sin bx - be^{ax} \cos bx}{b^2 + a^2}}$$

③  $\int \frac{p(x)}{q(x)} dx$   $q(x) = (x-a_1)^{k_1} \dots (x^2+bx+c)^{l_1}$

$$= \frac{A_1}{(x-a_1)} + \frac{A_2}{(x-a_1)^2} \dots + \frac{A_{k_1}}{(x-a_1)^{k_1}} + \dots + \frac{B_1x+C_1}{x^2+bx+c} + \frac{B_lx+C_l}{(x^2+bx+c)^{l_1}} + C$$

a)  $\int \frac{x^2+x+1}{x^3+2x^2+x} dx = \int \frac{x^2+x+1}{x(x+1)^2} dx = \int \left( \frac{A_1}{x} + \frac{A_2}{x+1} + \frac{A_3}{(x+1)^2} \right) dx$

$$A_1(x+1)^2 + A_2x(x+1) + A_3x = x^2+x+1$$

$$A_1 + A_2 = 1$$

$$2A_1 + A_2 + A_3 = 1$$

$$A_1 = 1$$

$$A_2 = 0$$

$$A_3 = -1$$

$$= \int \frac{1}{x} dx - \int \frac{1}{(x+1)^2} dx = \ln|x| + \frac{1}{x+1} + C$$

$$x+1 = t$$

$$dx = dt$$

$$b) \int \frac{x^3 + 4x^2 + 25x - 25}{x^4 + 25x^2} dx = \int \frac{x^3 + 4x^2 + 25x - 25}{x^2(x^2 + 25)} dx =$$

$$= \int \left( \frac{A_1}{x} + \frac{A_2}{x^2} + \frac{K_1x + N_1}{x^2 + 25} \right) dx =$$


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$$A_1x(x^2 + 25) + A_2(x^2 + 25) + (K_1x + N_1)(x^2) = x^3 + 4x^2 + 25x - 25$$

$$A_1 + K_1 = 1$$

$$A_1 = 1$$

$$K_1 = 0$$

$$A_2 + N_1 = 4$$

$$A_2 = -1$$

$$N_1 = 5$$

$$25A_1 = 25$$

$$25A_2 = -25$$


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$$= \int \frac{1}{x} dx - \int \frac{1}{x^2} dx + 5 \int \frac{1}{x^2 + 25} dx =$$

$$= \ln|x| + \frac{1}{x} + \frac{5 \cdot 5}{25} \arctg \frac{x}{5} + C = \boxed{\ln|x| + \frac{1}{x} + \arctg \frac{x}{5} + C}$$

$$\frac{x}{5} = t$$

$$\frac{1}{5} dx = dt$$

c)  $\int \frac{2}{(x^2+1)^2} dx = A \ln(x^2+1) + B \operatorname{arctg}\left(\frac{2x}{x^2+1}\right) + \frac{Kx+N}{x^2+1} + C$

$A=0$        $K=B=1$   
 $N=0$

$= \operatorname{arctg} x + \frac{x}{x^2+1} + C$

4.

a)  $\int \frac{1}{5+4\cos x} dx =$

$= \int \frac{2 dt}{5+4\left(\frac{1-t^2}{1+t^2}\right)(1+t^2)} =$

$= \int \frac{2 dt}{5+5t^2+4-4t^2} = 2 \int \frac{dt}{9(1+(\frac{t}{3})^2)} = \frac{2}{3} \operatorname{arctg}\left(\frac{t}{3}\right) + C$

$$t = \operatorname{tg} \frac{x}{2}$$

$$dx = 2 \frac{dt}{1+t^2}$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\cos x = \frac{1-t^2}{1+t^2}$$

b)  $\int \frac{1}{\sin^2 x + 2\cos^2 x} dx = \int \frac{1}{1+\cos^2 x} dx =$

$= \int \frac{dt}{(1+t^2)(1+\frac{1}{1+t^2})} =$

$= \int \frac{dt}{1+t^2+1} = \int \frac{dt}{2+t^2} = \frac{1}{2} \int \frac{\sqrt{2} du}{1+u^2} =$

$$\frac{t}{\sqrt{2}} = u$$

$$\frac{1}{\sqrt{2}} dt = du$$

$$\frac{\sqrt{2}}{2} \operatorname{arctg} \frac{\operatorname{tg} x}{\sqrt{2}} + C$$

$$t = \operatorname{tg} x$$

$$t^2 = \operatorname{tg}^2 x = \frac{1}{\cos^2 x} - 1$$

$$\cos^2 x = \frac{1}{1+t^2}$$

$$dt = \frac{1}{\cos^2 x} dx$$

$$dx = \frac{dt}{1+t^2}$$

